

Linear regression (simple and inference)

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Simple Linear Regression: The Foundations

Statistical Foundations

Inference in Simple Linear Regression



Predicting body fat percentage

What lifestyle factors do you think are predictive of body fat percentage?



Example: Predicting body fat percentage

Problem: Predict body fat percentage based on observed lifestyle characteristics

Solution:

- Look at data where we know body fat percentage and some observed characteristics
- Build a decision rule that predicts body fat percentage as a function of the observed characteristics

To keep things simple, let's focus only on **hours watched on Netflix per week**.

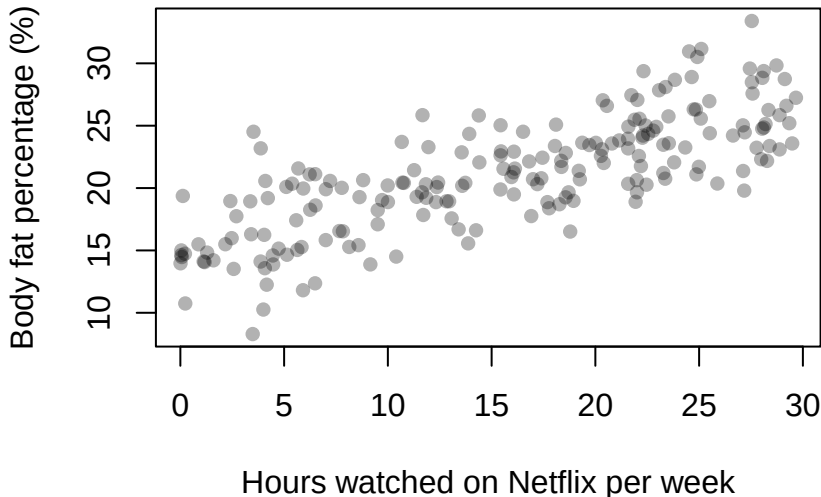


Setting up the problem

- The value we seek to predict is the **response** (or dependent/output) variable:
 - Y = body fat percentage (%)
- The variable we use to guide prediction is the **predictor** (or explanatory/input) variable:
 - X = hours watched on Netflix per week



Explore the data: scatterplot



View the data as points in the $X \times Y$ plane. There seems to be a **linear** relationship!



The regression model

Y = response or outcome variable

X = explanatory or predictor variable

A **linear** relationship is written:

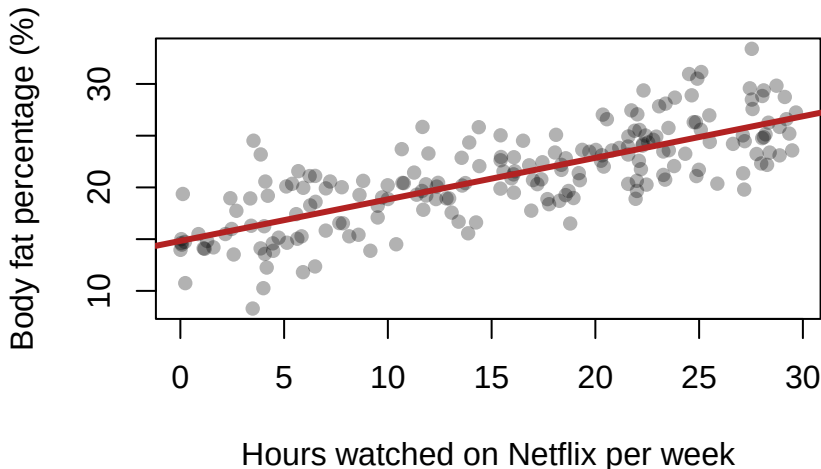
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- β_0 is the **intercept** (in units of Y)
- β_1 is the **slope** (in units of Y per unit of X)
- ε is the **error** or residual



Linear prediction

Recall the equation of a line: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$



As Netflix hours go up, body fat percentage tends to go up. But how do we find the “best” line?



Fitted values and residuals

- The **fitted value** $\hat{Y}_i$ is our prediction for observation i :

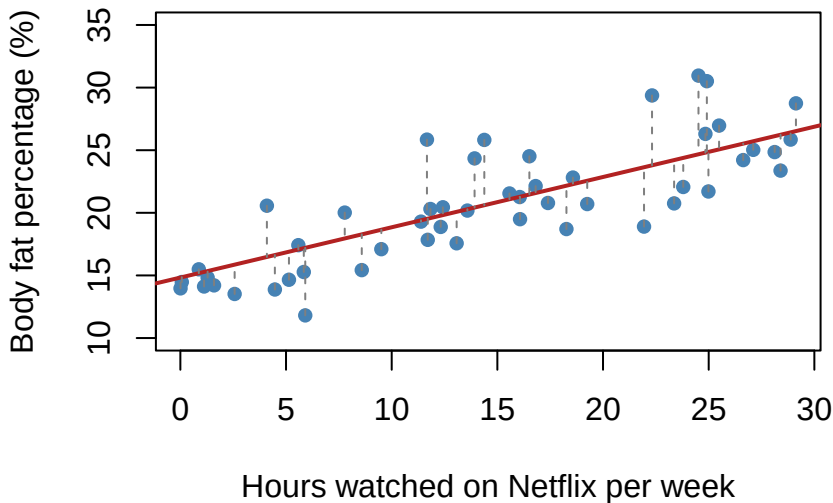
$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

- The **residual** e_i is how wrong our prediction is:

$$e_i = Y_i - \hat{Y}_i$$

- We can write: $Y_i = \hat{Y}_i + e_i$ (observed = predicted + error)

Visualizing residuals



The dashed lines are the residuals: $e_i = Y_i - \hat{Y}_i$



Least squares

How do we find the “best” line?

Ideally, we want to minimize the size of all residuals:

- If they were all zero, we'd have a perfect fit
- But there's a trade-off: moving closer to some points means moving away from others



Least squares

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Least Squares: Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize $\sum_{i=1}^n e_i^2$

$$\sum_{i=1}^n e_i^2 = (Y_1 - \hat{Y}_1)^2 + (Y_2 - \hat{Y}_2)^2 + \cdots + (Y_n - \hat{Y}_n)^2$$



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- Squaring makes all errors positive (can't cancel out)
- Squaring penalizes large errors more heavily
- Leads to nice mathematical properties (unique solution)
- The formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ are:

$$\hat{\beta}_1 = r_{xy} \times \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

where r_{xy} is the correlation and s_x , s_y are standard deviations



Least squares in R

```
model <- lm(Body_Fat ~ Netflix_Hours, data=netflix_data)
summary(model)
```

Call:

```
lm(formula = Body_Fat ~ Netflix_Hours, data = netflix_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.9344	-1.8695	0.0093	1.8318	8.2710

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.82928	0.41936	35.36	<2e-16 ***
Netflix_Hours	0.40153	0.02338	17.18	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.889 on 198 degrees of freedom

Multiple R-squared: 0.5984, Adjusted R-squared: 0.5964

F-statistic: 295 on 1 and 198 DF, p-value: < 2.2e-16



Interpreting the output

Our prediction equation is:

$$\widehat{\text{Body Fat}} = 14.83 + 0.4 \times \text{Netflix Hours}$$

- **Intercept** ($\hat{\beta}_0 = 14.83$): Predicted body fat percentage when Netflix Hours = 0.



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- **Intercept** ($\hat{\beta}_0 = 14.83$): Predicted body fat percentage when Netflix Hours = 0.
- **Slope** ($\hat{\beta}_1 = 0.4$): Each additional hour of Netflix per week is associated with a 0.4 percentage point increase in predicted body fat.



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- **Intercept** ($\hat{\beta}_0 = 14.83$): Predicted body fat percentage when Netflix Hours = 0.
- **Slope** ($\hat{\beta}_1 = 0.4$): Each additional hour of Netflix per week is associated with a 0.4 percentage point increase in predicted body fat.
- A 10 hour increase $\rightarrow$ 4 percentage point increase in predicted body fat

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Understanding the relationship: Covariance

To understand how we get the slope and intercept, we need to understand how X and Y vary together.

- **Variance** measures how one variable spreads out:

$$s_y^2 = \frac{1}{n-1} \sum (Y_i - \bar{Y})^2$$

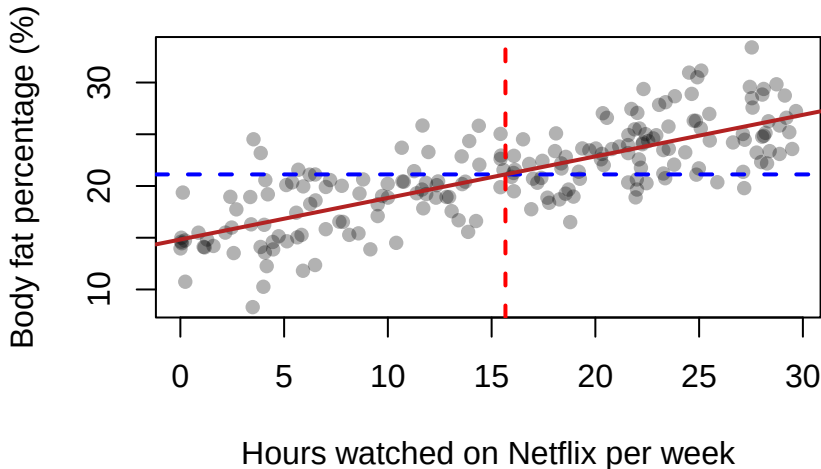
- **Covariance** measures how two variables vary *together*:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- Positive covariance: When X is above its mean, Y tends to be above its mean
- Negative covariance: When X is above its mean, Y tends to be below its mean
- Zero covariance: No linear relationship



Visualizing covariance



The covariance measures how points fall relative to the mean lines. Points in quadrants I and III contribute positively; points in II and IV contribute negatively.



A standardized measure: Correlation

Covariance is hard to interpret because it depends on the units of measurement.

Correlation standardizes the covariance:

$$r = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{s_x s_y} = \frac{\text{cov}(X, Y)}{\sqrt{s_x^2 s_y^2}}$$



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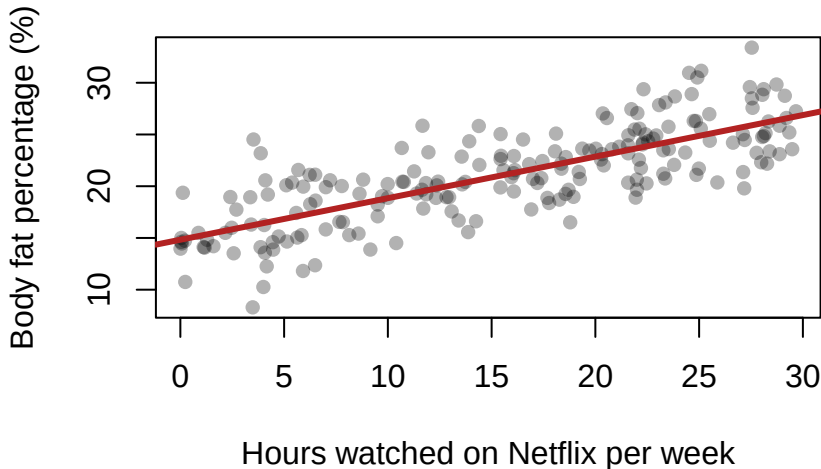
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- Always between -1 and $+1$
- $r > 0$: positive linear relationship
- $r < 0$: negative linear relationship
- $r = 0$: no linear relationship
- $|r| = 1$: perfect linear relationship



Correlation for our Netflix example

```
cor(netflix_data$Netflix_Hours, netflix_data$Body_Fat)  
[1] 0.7735597
```





Connection to least squares

Remember the formulas for the slope and intercept:

$$\hat{\beta}_1 = r_{xy} \times \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$



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- It also depends on the ratio of standard deviations
- If correlation is zero, the slope is zero (no linear relationship)
- The intercept ensures the regression line passes through $(\bar{X}, \bar{Y})$



How good is our model?

Two key questions:

1. How much of the variation in Y does our model explain?
2. How wrong are our predictions, on average?

Decomposing the variance

Remember: $Y = \hat{Y} + e$ (observed = predicted + residual)

The total variation in Y can be split:

$$\underbrace{\sum (Y_i - \bar{Y})^2}_{\text{SST (Total)}} = \underbrace{\sum (\hat{Y}_i - \bar{Y})^2}_{\text{SSR (Regression)}} + \underbrace{\sum e_i^2}_{\text{SSE (Error)}}$$

- **SST**: Total variation in Y
- **SSR**: Variation explained by the regression
- **SSE**: Variation left unexplained (residuals)



R^2 : Coefficient of determination

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

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- For our body fat model: $R^2 = 0.598$
- About 59.8% of the variation in body fat percentage is explained by Netflix hours
- Note: For simple regression, $R^2 = r^2$ (correlation squared)



Residual standard error

$$s_e = \sqrt{\frac{\text{SSE}}{n-2}} = \sqrt{\frac{\sum e_i^2}{n-2}}$$

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- For our model: $s_e = 2.89$
- On average, our predictions are off by about 2.89 percentage points
- Why $n - 2$? We “used up” 2 degrees of freedom estimating $\hat{\beta}_0$ and $\hat{\beta}_1$



The probability model

The **simple linear regression model** assumes:

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

- $\beta_0 + \beta_1 X$ is the “true line” (systematic part)



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- Errors are Normally distributed with mean 0 and constant variance
- This means: $Y|X \sim N(\beta_0 + \beta_1 X, \sigma^2)$



Why does this matter?

The probability model allows us to:

- Construct **confidence intervals** for the slope and intercept



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Let's see these ideas in action with a new example!

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What personal characteristics about an instructor do you think are predictive of the scores they receive on student evaluations?



Economics of Education Review

Volume 24, Issue 4, August 2005, Pages 369–376



Beauty in the classroom: instructors' pulchritude and putative pedagogical productivity

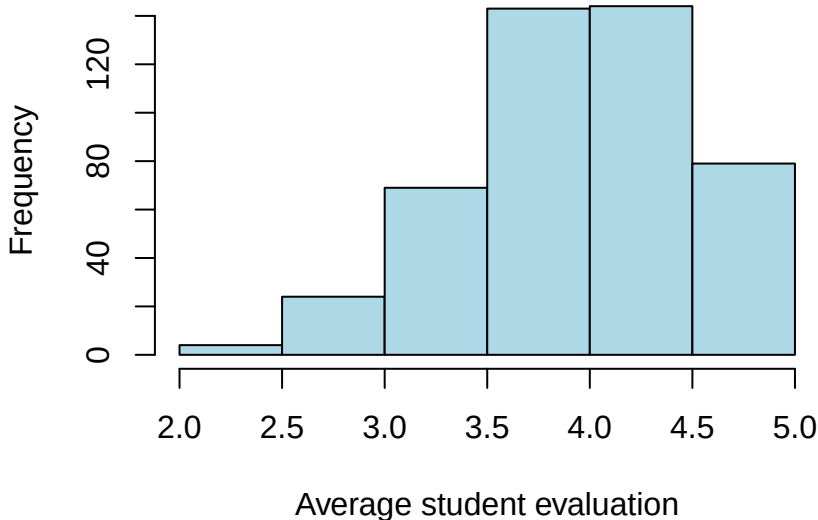
Daniel S. Hamermesh  , Amy Parker



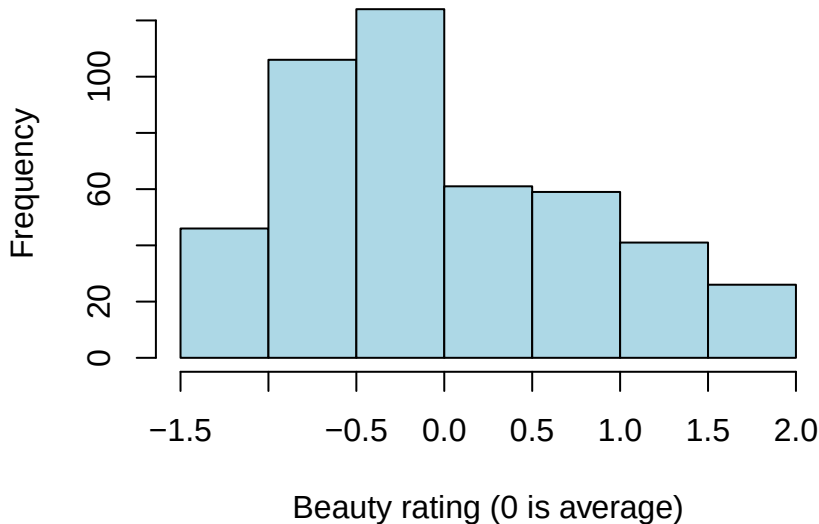
Hamermesh & Parker (2005) data set

- Student evaluations of $N = 463$ instructors at UT Austin, 2000-2002
- For each instructor:
 - **eval**: average student evaluation of teacher
 - **beauty**: average beauty score from a six-student panel
 - **gender**: male or female
 - **credits**: single- or multi-credit course
 - **age**: age of instructor
 - (and more...)

Explore the data: eval

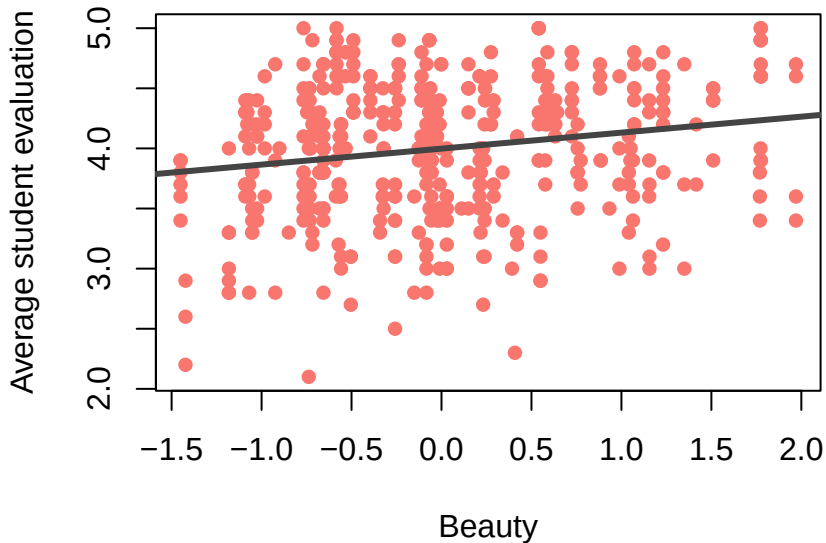


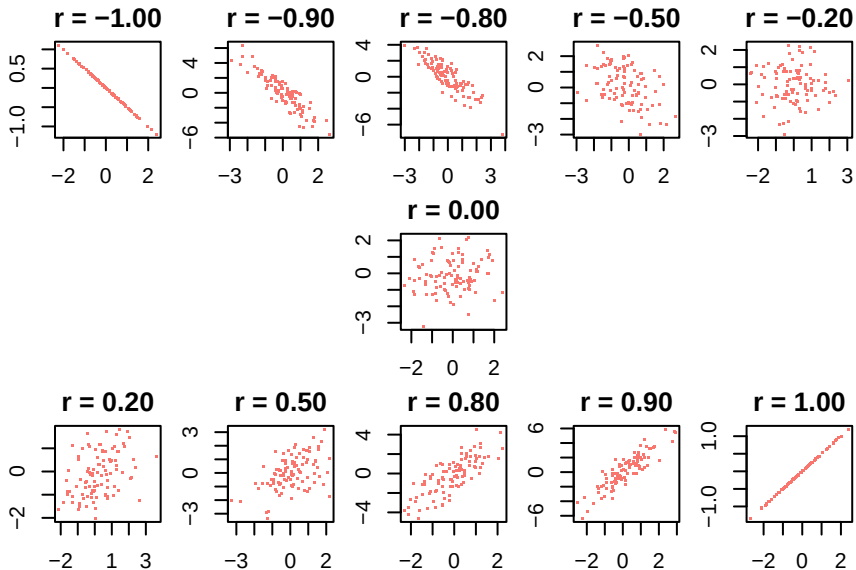
Explore the data: beauty



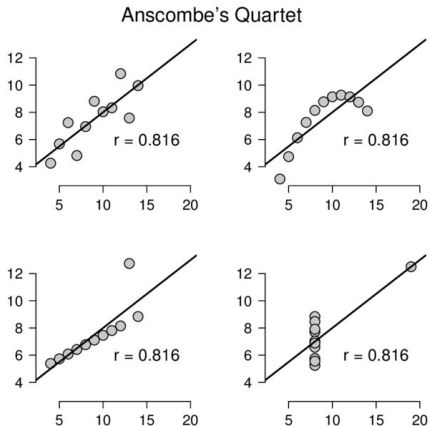
Do you think there is a positive or negative relationship between beauty and teaching ratings?

Explore the data





Correlation



Remember, correlation measures *linear* relationships and can be sensitive to outliers. Plot your data! (image from Forstmann & Wagenmakers 2015)



Correlation between eval and beauty

```
cor(profs$eval, profs$beauty)
```

```
[1] 0.189
```

How can we interpret this?



Let's build a simple regression model

```
model <- lm(eval ~ beauty, data=profs)
summary(model)
```

```
Call:
lm(formula = eval ~ beauty, data = profs)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.8002 -0.3630  0.0725  0.4021  1.1037
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.9983     0.0253  157.73  < 2e-16 ***
beauty         0.1330     0.0322    4.13  4.2e-05 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.545 on 461 degrees of freedom
Multiple R-squared:  0.0357, Adjusted R-squared:  0.0336
F-statistic: 17.1 on 1 and 461 DF,  p-value: 4.25e-05
```



Interpreting the model

- `eval` is the **response** variable (Y); `beauty` is the **predictor** variable (X)



Interpreting the model

- eval is the **response** variable (Y); beauty is the **predictor** variable (X)
- Simple regression uses the best fit line to give us a linear equation to predict Y from X :

$$\widehat{\text{eval}} = 3.998 + 0.133 \cdot \text{beauty}$$



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- We can predict the evaluation score for someone based on their beauty score. What is the predicted evaluation score for someone with a beauty score of 1?



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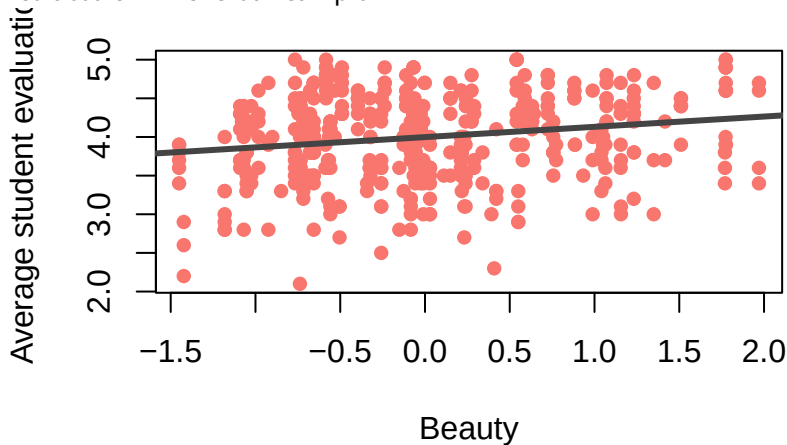
$$\widehat{\text{eval}} = 3.998 + 0.133 \cdot \text{beauty}$$

- We can predict the evaluation score for someone based on their beauty score. What is the predicted evaluation score for someone with a beauty score of 1?
- What do the coefficients mean?



How wrong is our estimate?

We're using this sample to learn about the whole population of UT instructors. This is our sample:

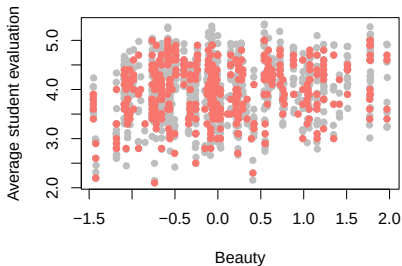


But what does the population that it was drawn from look like?

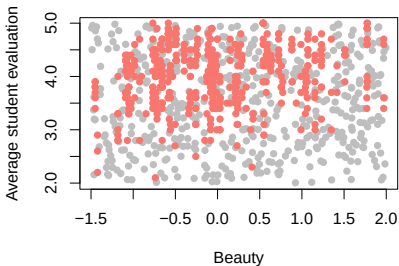


Statistical significance of the model

Is this the population?



Or is this the population?





Confidence intervals & statistical significance

- The population regression line (the best fit line in the population) is $Y = \beta_0 + \beta_1 X$ (we can't know this)



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- The sample intercept and slope $\hat{\beta}_0$ and $\hat{\beta}_1$ are our best estimates for the population intercept and slope β_0 and β_1



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- Our regression equation is the best fit line in the sample, or $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ (this is what we get from our sample data)
- The sample intercept and slope $\hat{\beta}_0$ and $\hat{\beta}_1$ are our best estimates for the population intercept and slope β_0 and β_1
- But we need to get a sense of how close $\hat{\beta}_0$ and $\hat{\beta}_1$ are to β_0 and β_1 !



Confidence intervals for the slope and intercept

We can use the model output to create a confidence interval for the slope (or the intercept):

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.9983	0.0253	157.73	< 2e-16	***
beauty	0.1330	0.0322	4.13	4.2e-05	***

- 95% CI for the slope is $\hat{\beta}_1 \pm t^*SE(\beta_1)$
(estimate $\pm$ (critical value)(standard error))



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(estimate $\pm$ (critical value)(standard error))
- t^* is the critical value from a t distribution with $n - 2$ degrees of freedom: $\text{qt}(0.975, 463-2) \approx 1.97$



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beauty	0.1330	0.0322	4.13	4.2e-05	***

- 95% CI for the slope is $\hat{\beta}_1 \pm t^* \text{SE}(\beta_1)$
(estimate $\pm$ (critical value)(standard error))
- t^* is the critical value from a t distribution with $n - 2$ degrees of freedom: $\text{qt}(0.975, 463-2) \approx 1.97$
- 95% CI for the slope is $0.133 \pm (1.97)(0.0322)$



Confidence intervals for the slope and intercept

We can use the model output to create a confidence interval for the slope (or the intercept):

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- Interpret this just like a bootstrapped CI!



Confidence intervals in general

Some rules of thumb:

- 95% CI for pretty much anything in this class (sample mean, intercept, slope, etc) is (estimate) $\pm 2(\textit{standard errors})$



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- Every estimator has a standard error, and every standard error has an associated estimator



Confidence intervals, the easy way

```
confint(model)

              2.5 % 97.5 %
(Intercept) 3.9485  4.048
beauty      0.0698  0.196
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- β_1 : We are 95% confident that the incremental impact of each additional beauty point is between 0.07 and 0.20 student evaluation points.



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- β_1 : We are 95% confident that the incremental impact of each additional beauty point is between 0.07 and 0.20 student evaluation points.
- β_0 : We are 95% confident that the average student evaluation score for average-looking professors ($\text{beauty} = 0$) is between 3.95 and 4.05.



Confidence intervals for predictions v.s. parameters

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95% confidence intervals for predictions (aka prediction intervals): The interval captures 95% of *future y values* for a given x



Testing and statistical significance

	Estimate	Std. Error	t value	Pr(> t)
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- If $\beta_1 = 0$, then we'd know that beauty has no impact on eval in the population



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- $\hat{\beta}_1 = 0.133 \neq 0$ in our sample, but was that just due to chance?
- We can answer this by testing the null hypothesis that $\beta_1 = 0$ against the alternative that $\beta_1 \neq 0$, accepting a false positive rate (type I error) of $\alpha = 0.05$



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- This is a conventional choice that balances being too strict (missing real effects) with being too lenient (claiming effects that don't exist)
- By setting α ahead of time, we control how often we make this mistake



Testing and statistical significance (continued)

	Estimate	Std. Error	t value	Pr(> t)
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- The p -value for the slope is 4.2×10^{-5} ; since it is very small, it is very unlikely that $\hat{\beta}_1$ is this big just by chance if β_1 is truly zero.



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- We **reject** the null hypothesis that $\beta_1 = 0$ at level $\alpha = 0.05$
- In other words, it does seem like there is a real (**statistically significant**) relationship between how good-looking professors are and their evaluation scores!



Statistical significance and confidence intervals

Some reminders about statistical significance

- You must pick a type I error ahead of time (without looking at the data)

Connection to confidence intervals: A 95% CI contains *all the values* for the parameter that the hypothesis test *fails to reject* with a type I error of 0.05

A confidence interval is a *set of plausible values* for the parameter – those which aren't ruled out by your data.



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Some reminders about statistical significance

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- The p-value is *not* the probability that the null hypothesis is true. It *assumes* the null hypothesis is true and measures how unlikely it would be to see results as extreme as ours due to chance

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- Look at the difference in $\widehat{\text{eval}}$ between someone super-hot (beauty = 2) and super-not (beauty = -1.5); the difference in their predicted student evaluations is $3.5 \cdot 0.133 = 0.466$

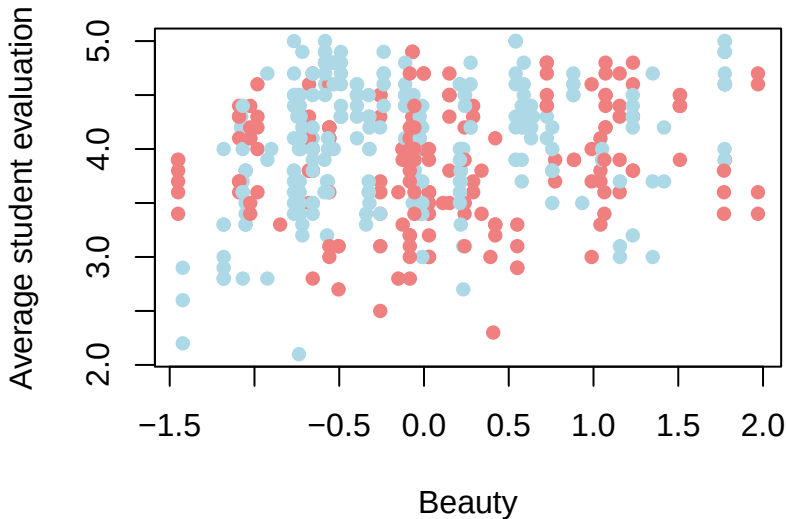
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(Spoiler alert: No.)



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